

Power BI Training – Introduction

◆ 1. What is Power BI?

Power BI is a Business Intelligence (BI) and data visualization tool developed by Microsoft.

It enables users to: Connect to multiple data sources (databases, Excel, APIs, cloud services).

Transform and clean data to prepare it for analysis.

Create data models with relationships and calculations.

Build interactive dashboards and reports with charts, KPIs, and slicers.

Share insights with others via the cloud (Power BI Service).

👉 In short: Power BI turns raw data into interactive insights to support decision-making.

◆ 2. Power BI Ecosystem (Components)

Power BI consists of several tools and services:

Power BI Desktop, Free Windows application.

Used by data analysts to design reports.

Includes: Power Query Editor → for data cleaning, Data Modeling → creating relationships and calculated measures, Report View → building charts and dashboards.

Power BI Service (Cloud)

Online SaaS platform (app.powerbi.com). Used to publish, share, and collaborate on reports.

Features: Dashboards (single-page views), Scheduled data refresh, Collaboration with colleagues.

Power BI Mobile: Apps for Android & iOS.

Used to access dashboards on-the-go.

 **Power BI Gateway:** Connects on-premises data sources (like SQL Server inside an office) with the Power BI cloud. Ensures data refresh works securely.

 **Power BI Report Server:** On-premise report server (for companies that don't use the cloud). Useful for highly secure environments.

Power BI Workflow :

The typical workflow in Power BI is:

1. Connect → Data Sources (Excel, SQL, Web, APIs)
2. Transform → Clean & Shape data using Power Query
3. Model → Build relationships between tables (Fact & Dimension)

4. Analyze → Use DAX to create calculated fields and measures
5. Visualize → Create charts, KPIs, dashboards
6. Publish → Share on Power BI Service
7. Collaborate → Teamwork, scheduled refresh, sharing insights

Power BI Features

✓ Data Connectivity: Connects with 100+ sources: Excel, SQL Server, Oracle, Azure, APIs, Salesforce, SharePoint, etc.

✓ Data Transformation (ETL): Power Query Editor allows cleaning & shaping data:

Remove nulls, Merge tables, Split columns, Filter rows

✓ Data Modeling: Build relationships between tables.

Example: Sales (fact) table linked with Products (dimension) table.

✓ DAX (Data Analysis Expressions): Formula language used in Power BI.

Example formulas:

SUM(Sales[Revenue]) → Total Revenue

CALCULATE(SUM(Sales[Revenue]), Region="Asia") → Revenue from Asia

✓ Visualizations

Bar, line, pie, treemap, map, gauge, KPI indicators, scatter, custom visuals.

✓ Dashboards

Combine multiple visuals into a single-page summary. Live and interactive.

✓ AI Features

Natural Language Q&A → Ask a question like “Show sales by region” and get a chart.

Data Source --- Power Query Editor --- Data Modelling --- Report Building ---- Power BI Services --- Mobile/web Apps

◆ 6. Benefits of Power BI

Easy to learn, especially for Excel users. Handles large datasets efficiently. Supports both self-service BI and enterprise BI. Real-time analytics with live dashboards. Cross-platform access (desktop, cloud, mobile).

Seamless integration with Microsoft products (Excel, Teams, Azure).

◆ 7. Example Use Cases

Coffee Sales Analysis , Blinkit Analysis , Heart Disease Analysis, Adventure Works Analysis

